SPECIFICATION — HARMONY PROFILE & PORTABILITY

PRIVACY-PRESERVING LONGITUDINAL COMPATIBILITY PROFILE, PORTABLE REPRESENTATION, AND THIRD-PARTY INTEGRATION FRAMEWORK (NON-LIMITING)

A. Technical Field

This disclosure relates to portable physiological profile systems and, more particularly, to a privacy-preserving longitudinal compatibility profile (hereinafter "Harmony Profile") derived from multi-node physiological sensing data, methods for representing and exporting the profile in portable formats, and frameworks for integrating the profile with third-party applications and platforms.

B. Background and Motivation (Non-Limiting)

Existing health and wellness platforms generate user profiles that are typically locked to a single application ecosystem, expressed in raw numeric formats, and lack portability. No known system generates a longitudinal compatibility profile from multi-node intimate physiology sensing data that can be exported, shared, or integrated with third-party platforms while maintaining zero-linkage privacy guarantees. The Harmony Profile addresses this by creating a versioned, portable, and privacy-preserving data construct that represents biological resonance characteristics without exposing raw physiological measurements.

C. Harmony Profile Definition and Architecture (Non-Limiting)

In certain embodiments, a Harmony Profile is a structured data construct derived from one or more multi-node physiological sensing sessions processed through the system data abstraction pipeline. The Harmony Profile comprises:

1. Pattern Signature Layer: A set of normalized, non-invertible pattern signatures representing characteristic physiological behavior across multiple sensing domains (e.g., rigidity dynamics, tissue compliance, pelvic activity, cardiovascular context).
2. Temporal Evolution Layer: A versioned history of pattern signatures tracking how the user's physiological characteristics change over time, stored as encrypted temporal segments.
3. Alignment Descriptor Layer: Pre-computed descriptors optimized for cross-user alignment scoring, enabling compatibility analysis without re-processing raw data.
4. Confidence and Coverage Metadata: Indicators of data quality, node coverage, temporal depth, and measurement confidence associated with each profile element.
5. Disclosure Control Layer: A permission matrix defining which profile elements may be shared, at what level of detail, and under what consent conditions.

D. Longitudinal Profile Evolution (Non-Limiting)

In various embodiments, the Harmony Profile is not a static snapshot but a living data construct that evolves as new physiological data is acquired. The profile evolution system supports:

- Incremental updates: New sensing sessions append to the temporal evolution layer without overwriting historical data
- Baseline drift detection: The system identifies and flags significant shifts in baseline physiological characteristics over weeks or months
- Version tagging: Each significant profile update generates a version identifier enabling point-in-time comparison
- Recovery trajectory tracking: Changes in profile characteristics following health interventions or lifestyle changes are tracked as named recovery trajectories
- Confidence growth: Profile confidence indicators improve as temporal depth and measurement coverage increase

E. Zero-Linkage Privacy Architecture (Non-Limiting)

In various embodiments, the Harmony Profile enforces zero-linkage integrity through the following mechanisms:

1. Non-Invertible Generation: Profile elements are generated through one-way transformations such that raw

sensor data, explicit anatomical measurements, or personally identifiable physiological characteristics cannot be reconstructed from the profile.

2. Abstracted Representation: All profile elements are expressed as abstract indices, pattern signatures, normalized vectors, or symbolic descriptors rather than explicit physiological values.

3. Layered Disclosure: The profile is structured in disclosure layers ranging from minimal summary indicators to detailed pattern descriptors. Each layer requires explicit user consent to share.

4. Screenshot Safety: Exported profile representations (including visual formats) contain no explicit anatomical data, raw measurements, or content that could identify specific physiological characteristics if captured or shared without authorization.

5. Encryption at Rest: All profile data is encrypted using user-controlled keys. The system does not retain decryption capability for profile data at rest.

F. Partial Disclosure and Progressive Sharing (Non-Limiting)

In certain embodiments, the Harmony Profile supports partial disclosure through a progressive sharing architecture:

1. Level 0 — Presence Only: A binary indicator that a Harmony Profile exists, without revealing any profile content.
2. Level 1 — Summary Card: A minimal, abstract representation (e.g., a symbolic visual or single composite index) suitable for display on third-party platforms.
3. Level 2 — Dimensional Overview: Per-dimension summary scores (e.g., timing, intensity, rhythm) without underlying pattern details.
4. Level 3 — Alignment Preview: Estimated compatibility indicators computed against another consenting user's profile at the same or lower disclosure level.
5. Level 4 — Full Pattern Comparison: Detailed cross-user alignment analysis including per-dimension sub-scores, temporal trends, and zone-based compatibility maps. Requires mutual Level 4 consent.

Disclosure levels are independently configurable per user. A user at Level 4 sharing with a user at Level 2 will produce a Level 2 comparison result. The system always operates at the lower of the two users' consent levels.

G. Harmony Cards — Portable Profile Representations (Non-Limiting)

In certain embodiments, the system generates exportable profile representations called Harmony Cards. A Harmony Card is a portable, screenshot-safe artifact that represents a subset of the user's Harmony Profile in a format suitable for sharing outside the Nexalis ecosystem. Harmony Cards may comprise:

- A symbolic visual representation (e.g., a Sigil or Lumen) derived from the user's profile
- A composite compatibility index or compatibility category
- A verification status indicating whether the profile data has been validated against device-derived measurements
- A timestamp or freshness indicator showing when the profile was last updated
- An optional QR code or deep link enabling another Nexalis user to initiate a consent-gated comparison

Harmony Cards contain no raw physiological data and are designed to convey profile characteristics entirely through abstract, symbolic, or categorical representations.

H. Third-Party Platform Integration Framework (Non-Limiting)

In various embodiments, the system provides integration mechanisms for third-party platforms. The integration framework operates in the following phases:

1. Phase 1 — Share Anywhere (Passive Display): Users export a Harmony Card as a static image, embeddable widget, or shareable link. Third-party platforms display the card without any data exchange or API integration. No platform-side integration is required.
2. Phase 2 — Verified Profile (Authenticated Display): Third-party platforms integrate a verification API to confirm that a displayed Harmony Card is authentic, current, and unmodified. The API returns only a verification status and timestamp, not profile data.
3. Phase 3 — Alignment API (Active Comparison): Third-party platforms integrate an alignment API that enables consent-gated compatibility comparisons between two Nexalis users within the third-party platform. The API accepts encrypted profile tokens and returns abstract alignment scores without exposing underlying profile data

to the third-party platform.

In all integration phases, the third-party platform never receives raw profile data, pattern signatures, or any information that could be used to reconstruct physiological measurements.

I. Match Typing and Preference Discovery (Non-Limiting)

In certain embodiments, the Harmony Profile enables two novel analytical capabilities:

1. Match Typing: Classification of compatibility patterns into categorical types based on objective physiological alignment data. Match types represent recurring patterns of cross-user compatibility observed across the user base (e.g., complementary rhythms, convergent baselines, or reciprocal response patterns). Match types are derived statistically and do not correspond to subjective preferences or self-reported characteristics.
2. Preference Discovery: Identification of compatibility characteristics that a user may not have been previously aware of. By analyzing cross-user alignment patterns across multiple comparisons, the system may surface previously unrecognized compatibility preferences expressed in abstract, non-explicit terms (e.g., "your profile shows strongest alignment with users who exhibit reciprocal timing patterns").

Both Match Typing and Preference Discovery operate on abstracted pattern signatures and produce only abstract, categorical, or symbolic outputs.

J. Profile Storage, Versioning, and Lifecycle (Non-Limiting)

In various embodiments, the Harmony Profile is stored and managed according to the following principles:

- User-Controlled Storage: The profile may be stored on the user's device, in encrypted cloud storage under user-controlled keys, or both
- Version History: All profile versions are retained to support longitudinal tracking and point-in-time comparison
- Selective Deletion: Users may delete specific profile versions, specific disclosure layers, or the entire profile at any time
- Export and Backup: Users may export their complete profile in an encrypted portable format for backup or migration
- Expiration and Freshness: Shared profile representations (Harmony Cards, alignment tokens) may include configurable expiration periods after which they are no longer valid for comparison

K. Exemplary Claim-Like Statements (Non-Limiting)

A provisional application does not require claims. The following statements are illustrative only and do not limit the disclosure.

1. A system comprising: a) a data abstraction pipeline configured to receive sensor data from one or more physiological sensing nodes associated with a user and generate a set of non-invertible pattern signatures; b) a profile generation module configured to assemble the pattern signatures into a structured Harmony Profile comprising a pattern signature layer, a temporal evolution layer, an alignment descriptor layer, and a disclosure control layer; and c) a profile export module configured to generate a portable, screenshot-safe Harmony Card representing a subset of the Harmony Profile; wherein the Harmony Profile and Harmony Card contain no raw sensor data and cannot be reverse-engineered to reconstruct physiological measurements.
2. The system of statement 1, wherein the temporal evolution layer stores versioned history of pattern signatures and tracks baseline drift, recovery trajectories, and confidence growth over time.
3. The system of statement 1, wherein the disclosure control layer defines multiple progressive disclosure levels, each requiring explicit user consent, and wherein cross-user comparisons operate at the lower of two users' consent levels.
4. The system of statement 1, further comprising a third-party integration framework operating in phases comprising passive display, authenticated verification, and consent-gated active comparison.
5. The system of statement 1, further comprising a match typing module configured to classify cross-user compatibility patterns into categorical types based on abstracted alignment data.
6. A method comprising: a) generating non-invertible pattern signatures from multi-node physiological sensing data; b) assembling a longitudinal Harmony Profile comprising versioned temporal segments; c) generating a portable Harmony Card containing a symbolic representation, a composite index, and a verification status; and d) sharing the Harmony Card via an export that contains no raw physiological data and is screenshot-safe.

7. The method of statement 6, further comprising: e) receiving a consent-gated comparison request from a second user; f) computing an alignment score at the lower of the two users' disclosed consent levels; and g) returning an abstract compatibility result without exposing either user's underlying pattern signatures to the other user.

8. The method of statement 6, further comprising performing preference discovery by analyzing cross-user alignment patterns across multiple comparisons and surfacing previously unrecognized compatibility characteristics in abstract, non-explicit terms.